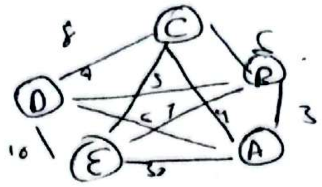
Muhammad Adil Saeed.

24K-0705.



Current	Frontier	Candidate	Visited
A	B, 3, C, 4 D, 6 E, 30	A→B A→C.	A.
A→B	C, 9, D. 8 E. 10.	A→B→C AB-D	AB A, C
A→C	B-10, D 12 C. 13.	"	A-C.
A→B→C.	D. 17, E 18	A→B→C→D	A, B, C
A→B→D	C, 16, E 18	A→B→D→C	A, B, D
A→B→C→D	E 27	A→B→D→C→E	A-B, C, D A, B, D, C
A→B→D→C	E, 26		
A→B→D→C→E	A, 30.		

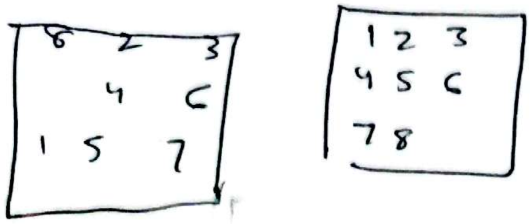
So path -

$A \xrightarrow{3} B \xrightarrow{5} D \xrightarrow{8} C \xrightarrow{9} E \xrightarrow{30} A.$

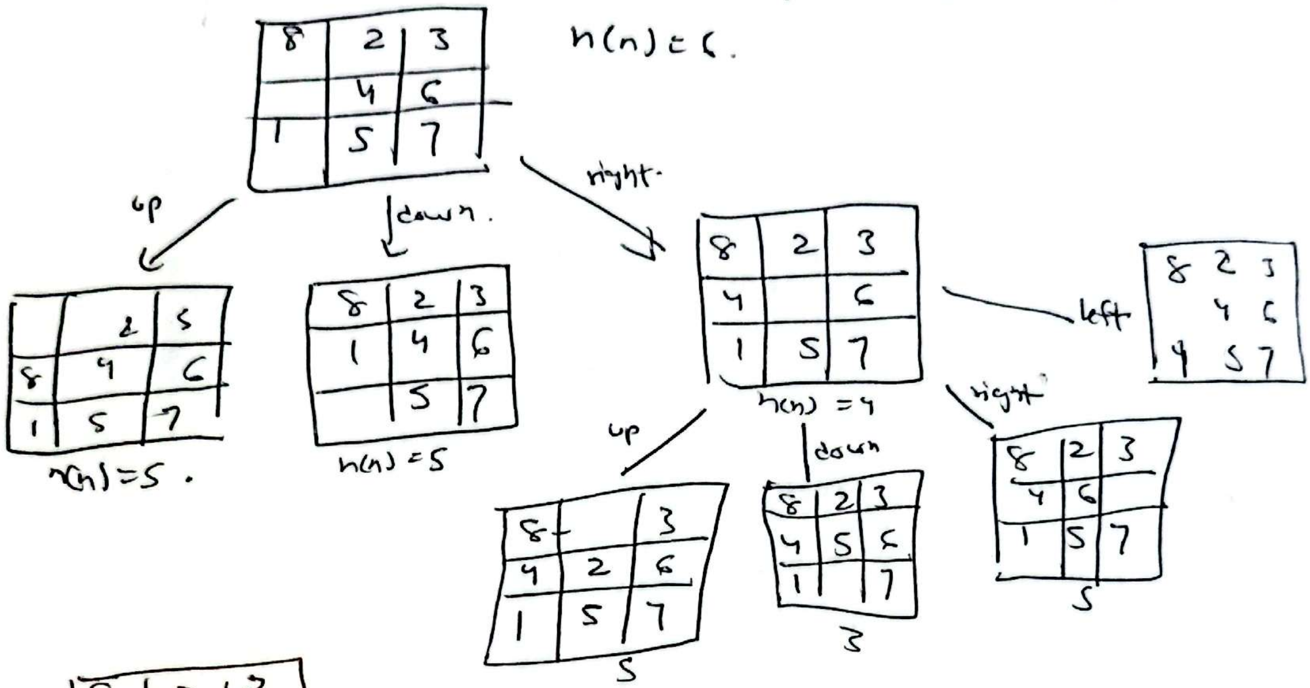
$= 55$.

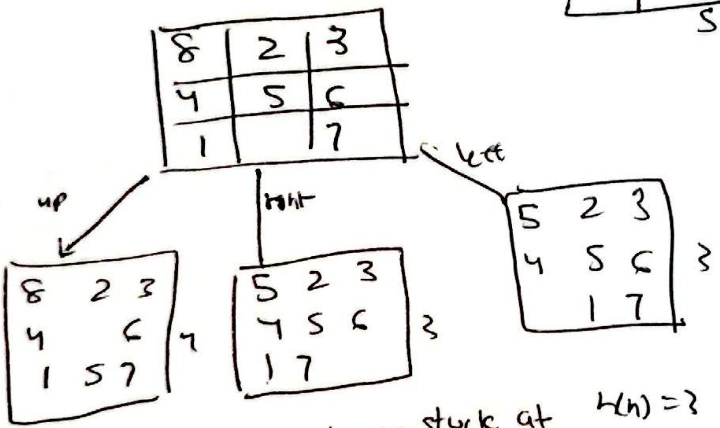
Scanned with CamScanner

Question 2 :



We will consider only up, down, left right moves





local maximum stuck at $h(n) = 3$

down → right → down → stuck

Scanned with CamScanner

Question 3.

initial

1 111010 – A   15+7+10+8=40

2 011101 – B   7 15+17+10 = 39.

3 100001 – C   15+17 = 22

4. 101101 – D   15 + 10+5 +17 = 47.

Crossover Di with A ; D with B.

101 101   101010 C_1
111 010   111101 C_2.

101 101   101 101 C_3
011 101   011 101 C_4.

111010 = 40
111101 = 54
10101 = 47
011 101 = 39.

B with C, B with A

111101   111 101
101 101   111 010.

111 101
101 101
111 010
111 101.

Mutation C on O_1 bit.

C_1 111 100
C_2 101 101
C_3 111 010
C_4 111 101.

A 111 100 = 32
B 101 101 = 47
C 111 010 = 40
D 111101 = 17

weight = 30

so bst 15.

101 101
1 sleeping bag
Pocket knife
torch,